

Power BI Student Dashboard Test Pack

This document is designed for trainers and instructors to evaluate students on real-world Power BI dashboard development skills using Product and Sales datasets.

Dataset Overview

File 1: Product.xlsx

Columns: ID, ProductName, ProductGroup, ProductCategory

File 2: SalesData.xlsx

Columns: OrderDate, OrderNumber, ProductKey, SalespersonKey, Salesperson, Supervisor, Manager, Channel, Quantity, UnitPrice

Task 1 – Data Import & Cleaning

1. Import both Excel files into Power BI.
2. Rename tables properly.
3. Verify data types.
4. Remove duplicates if required.
5. Create relationship between Product[ID] and SalesData[ProductKey].
6. Create Date hierarchy.

Executive Sales Dashboard

KPI Cards:

- Total Sales
- Total Quantity Sold
- Total Orders
- Average Order Value
- Total Products Sold

Charts:

- Monthly Sales Trend
- Top Product Groups by Sales
- Sales by Channel
- Product Category TreeMap
- Manager-Supervisor-Salesperson Matrix

Salesperson Performance Dashboard

Required Visuals:

- Salesperson vs Sales
- Monthly Sales by Channel
- Quantity by Product Category
- Ribbon Chart for Managers
- Scatter Chart for Quantity vs Sales

Product Analysis Dashboard

- Top 20 Products by Sales
- Bottom 10 Products by Sales
- Product Category Contribution %

- Heatmap: Product Group vs Channel
- Waterfall Chart for Monthly Sales Growth

Important DAX Measures

Total Sales

SUMX(SalesData, SalesData[Quantity] * SalesData[UnitPrice])

Total Orders

DISTINCTCOUNT(SalesData[OrderNumber])

Average Order Value

DIVIDE([Total Sales], [Total Orders])

Beginner Interview Questions

1. Difference between measure and calculated column?
2. What is Star Schema?
3. What is slicer?
4. Difference between SUM and SUMX?
5. What is Power Query?

Intermediate Questions

1. Create YTD Sales measure.
2. Explain row context.
3. Difference between ALL and ALLEXCEPT.
4. Explain inactive relationships.
5. Create rolling 3-month sales.

Advanced Questions

1. Explain CALCULATE deeply.
2. What is context transition?
3. Explain virtual tables.
4. How incremental refresh works?
5. How to optimize dashboard performance?

Practical Evaluation Criteria

Data Cleaning – 10 Marks
Data Modeling – 10 Marks
DAX Measures – 20 Marks
Visualization – 20 Marks
Dashboard Design – 15 Marks
Interactivity – 10 Marks
Performance Optimization – 5 Marks
Presentation – 10 Marks

Bonus Tasks

- Add Tooltips
- Add Drill-through
- Create Dark Theme Dashboard
- Add Mobile Layout

- Add AI Visual Insights

Dummy Dashboard Layout

KPI 1	KPI 2	KPI 3	KPI 4	KPI 5
Sales Trend Line Chart				
Bar Chart		Pie Chart		
TreeMap		Matrix Table		
Slicers Panel				

Prepared for Power BI practical training and student evaluation.